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Programming with
JAVA

Unit -3
(Part -3)
Topic – Inner Class

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Prepared By:
Prof. Pratik B. Soni
(Asst. Prof. – I.T. Dept, MBIT)

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Inner classes

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Inner classes - Basics

- Inner classes ===== Nested classes
- Java inner class or nested class is a class that is declared inside the class or interface.
- Mainly used for logically group classes and interfaces in one place to be more readable and maintainable.

```
class OuterClass
{
    //code
        class InnerClass
        {
                //code
        }
}
```

- Additionally, it can access all the members of the outer class, including private data members and methods.

Inner classes vs nested classes

- There are two types of nested classes **non-static and static** nested classes.
- ***The non-static nested classes are also known as inner classes.***
- **Non-static nested class (inner class – nested class)**
  - Member inner class
  - Anonymous inner class
  - Local inner class
- **Static nested class (only nested class)**

Inner classes - Basics

Inner classes / interfaces	**Notes**
Member Inner Class	A class created **within class** and **outside method**.
Anonymous Inner Class	A **class created for implementing an interface** or **extending class**. The java compiler decides its name.
Local Inner Class	A class was **created within the method**.
Static Nested Class	A **static class** was created within the class.
Nested Interface	An **interface created** within class or interface.

Inner classes - Advantages

1) Nested classes represent a particular **type of relationship that is it can access all the members** (data members and methods) of the outer class, including private.

2) Nested classes are used to develop **more readable and maintainable** code because it logically group classes and interfaces in one place only.

3) Code Optimization: **It requires less code to write**.

Inner classes - Examples

Member Inner Class

A class created **within class** and **outside method**.

```
class Outer1{
        private int data=130;
        class Inner1{
                void msg()
                {
                        System.out.println("data : "+data);
                }
        }
        public static void main(String args[])
        {
        Outer1 ot1=new Outer1();
        Outer1.Inner1 in1=ot1.new Inner1();
         in1.msg();
        }
}
```

OUTPUT

data : 130

Inner classes - Examples

Anonymous Inner Class

A **class created for implementing an interface** or **extending class**. The java compiler decides its name.

```
abstract class Person
{
 abstract void eat();
}
class AnonymousInner1
{
        public static void main(String args[])
        {
        Person p=new Person() {
           void eat() {
               System.out.println("anonymous");
                  }
          };
        p.eat();
        }
}
```

A class is created, **but its name is decided by the compiler**, which **extends the Person class** and provides the implementation of the **eat() method**.

An **object of the Anonymous class** is created that is **referred to by 'p'**, a reference variable of Person type.

OUTPUT

anonymous

Inner classes - Examples

Anonymous Inner Class

A **class created for implementing an interface** or **extending class**. The java compiler decides its name.

```
interface food
{
        void eat();
}
class AnonymousInner1
{
        public static void main(String args[])
        {
        food p=new food() {
            void eat() {
                System.out.println("anonymous");
                }
            };
        p.eat();
        }
}
```

OUTPUT

anonymous

Inner classes - Examples

Local Inner Class

A class was **created within the method**.
Object of Inner class is created in method.

```
public class localInner1
{
        private int data=30;//instance variable
        void display()
        {
                class Local{
                        void msg()
                        {   System.out.println(data);   }
                }
                Local l=new Local();
                l.msg();
        }
        public static void main(String args[])
        {
                localInner1 obj=new localInner1();
                obj.display();
        }
}
```

OUTPUT

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Inner classes - Examples

Static Nested Class

A **static class** was created within the class.

```
class Outer1
{
        static int data=30;
        static class Inner
        {
            void msg()
                { System.out.println(data); }
        }

        public static void main(String args[])
        {
                Outer1.Inner obj=new Outer1.Inner();
                obj.msg();
        }
}
```

OUTPUT

30

Inner classes - Examples

Nested Interface

An **interface created** within class or interface.

```
interface outer1
{
 void show();
        interface inner1
        {
        void msg();
        }
}
class Main1 implements outer1.inner1
{
   public void msg()
        { System.out.println("Hello"); }
  public static void main(String args[])
  {
        outer1.inner1 obj=new Main1();    //upcasting here
        obj.msg();
        }
}
```

OUTPUT

Hello